WINTER INTERNSHIP REPORT

Duration: 22 days (15th December, 2025 to 5th January, 2026)

"Emerson DCS system at the CDU unit of Guwahati Refinery, IOCL"

Submitted to:
Mr. Aman Chandra
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Submitted by:
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CERTIFICATE

This is to certify that Ms.**Indrani Chakraborty**, a student of **Electronics and Instrumentation Engineering** at **National Institute of Technology, Agartala** has successfully completed a 22 - days Winter internship at **Indian Oil Corporation Limited (IOCL), Guwahati Refinery** from **15-12-25** to 0**5-01-26.**

During the course of the internship, she was assigned to the Instrumentation Department, where she actively participated in gaining hands-on exposure to various industrial automation systems. The primary focus was on the Emerson DeltaV Distributed Control System (DCS) implemented in the Crude Distillation Unit (CDU).

The work has been satisfactory and approved for fulfillment of the training.

Mr. Aman Chandra
Instrumentation Department
Indian Oil Corporation Limited
Guwahati Refinery

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who contributed to making my 22-days Winter internship at **Guwahati Refinery, Indian Oil Corporation Limited (IOCL)**, a truly enriching and memorable experience. This internship marked a significant milestone in my academic and professional journey.

I am especially grateful to **Mr. Aman Chandra**, **Instrumentation Department**, for his exceptional mentorship and constant support. His guidance played a key role in enhancing my technical knowledge and practical understanding of industrial instrumentation.

I also thank the **Learning and Development Centre** of **Guwahati Refinery** for organizing the internship seamlessly and ensuring our awareness of safety protocols and operational procedures. Their efforts provided us with structured exposure to the refinery's automation systems.

My sincere appreciation goes to **National Institute of Technology, Agartala** for their support in facilitating this opportunity and encouraging industry-academic integration.

During the internship, exploring the **Emerson DeltaV Distributed Control System (DCS)** in the **Crude Distillation Unit (CDU)** deepened my understanding of real-world applications of process control.

Lastly, I thank the entire team at **Guwahati Refinery** for their warm hospitality and collaborative spirit. The skills and insights I've gained will undoubtedly shape my future endeavors in instrumentation engineering.

ABSTRACT

This report presents an in-depth account of the winter internship conducted **at Indian Oil Corporation Limited (IOCL), Guwahati Refinery**, focusing on the **Emerson DeltaV Distributed Control System (DCS)** implemented in the **Crude Distillation Unit (CDU).** The CDU is the primary unit for separating crude oil into useful fractions.

The report explores the DCS architecture, field instrumentation, signal transmission, I/O modules, and the role of Human-Machine Interface (HMI) in real-time monitoring. It emphasized system reliability through communication protocols like HART, Foundation Fieldbus, and Modbus and the importance of automation in ensuring safe and efficient refinery processes.

This exposure bridged theoretical knowledge with real-world applications, reinforcing the importance of instrumentation, control strategy design, and operational safety in the oil and gas sector.

CONCLUSION

The 22-days industrial training program conducted at **IOCL's Guwahati Refinery** offered participants a profound understanding of refinery operations, particularly through the lens of automation and process control. The program's core focus on the **Emerson Distributed Control System (DCS)** installed at the **Crude Distillation Unit (CDU)** allowed trainees to explore both the theoretical and practical aspects of modern control systems in a live industrial environment.

Throughout the training, participants gained firsthand knowledge of how critical data is transmitted from **field instruments**—often installed in remote and challenging locations—to the **central control room**. They observed how this data, in the form of analog and digital signals, is routed through **interfacing devices**, processed by **I/O modules** such as AI, DI, AO, and DO cards, and interpreted by the DCS controller to maintain precise process control.

The exposure to **signal transmission standards**, **two-wire systems**, **HART communication**, and the structured architecture of Emerson's DeltaV system helped clarify the complex pathways through which instrumentation supports real-time monitoring, alarm handling, and automated operations. In addition, participants closely studied how the **Human-Machine Interface (HMI)** empowers operators to interact with the system intuitively and effectively.

This immersive experience not only enhanced technical proficiency in **instrumentation and control** but also emphasized the role of **safety**, **redundancy**, and **system diagnostics** in ensuring reliability in refinery operations. The program reinforced the idea that DCS systems are not just control platforms, but essential enablers of operational excellence in the oil and gas industry.

Engaging interactions with experienced engineers and guided walkthroughs of live systems deepened learning and sparked enthusiasm for future industrial innovation. The participants leave the program with greater confidence, practical insight, and a sincere appreciation for the technologies and people that drive one of India's most critical industries.